

MODULE 1: INTRODUCTION

Contents

1. Intelligence	1
2. Artificial Intelligence (AI):	1
3. Key Concepts in AI	2
4. Benefits of AI.....	2
5. Limitations of AI.....	2
6. Classification of AI	2
6.1. Based on Functionality.....	2
6.2. Based on Capabilities.....	3
6.3. Based on Technology	3
6.4. Based on Applications.....	4
7. Agent.....	4
8. Environment	5
9. Key Concepts.....	7
10. PEAS (Performance measure, Environment, Actuators, Sensor)	7
11. Uncertainty in Artificial Intelligence.....	9

1. Intelligence: is the ability to learn, understand, and apply knowledge, solve problems, and adapt to new situations. It encompasses various cognitive abilities like reasoning, problem-solving, abstract thinking, and learning. Types of Intelligence:

- **Natural Intelligence:** This is the intelligence observed in humans and animals, allowing them to process information, learn from experiences, and adapt to their environment.
- **Emotional Intelligence:** The ability to recognize, understand, and manage one's emotions and the emotions of others.
- **Social Intelligence:** The capacity to interact effectively with others and understand social dynamics.
- **Logical-Mathematical Intelligence:** The ability to analyse problems logically, carry out mathematical operations, and investigate issues scientifically.

2. Artificial Intelligence (AI): refers to the simulation of human intelligence in machines that are programmed to think and learn like humans. AI systems are designed to perform tasks that typically require human intelligence, such as visual perception, speech recognition, decision-making, and language translation.

3. Key Concepts in AI

1. **Machine Learning:** A subset of AI that involves training machines to learn from data and improve their performance over time without being explicitly programmed.
2. **Neural Networks:** Computational models inspired by the human brain's network of neurons, used in deep learning to recognize patterns and make decisions.
3. **Natural Language Processing (NLP):** The ability of AI to understand, interpret, and generate human language, allowing for interactions between computers and humans in a natural way.
4. **Robotics:** The branch of AI that involves the design, construction, and operation of robots, which can perform tasks in the real world.
5. **Computer Vision:** The field that enables machines to interpret and understand visual information from the world, like human vision.
6. **Expert Systems:** AI systems that mimic the decision-making abilities of a human expert. They use a knowledge base of facts and rules to solve complex problems.
7. **Fuzzy Logic:** A form of logic used in AI that allows for reasoning with uncertain or imprecise information, like human reasoning.

4. Benefits of AI

1. **Automation:** AI automates repetitive tasks, improving efficiency.
2. **Decision-Making:** Analyses large datasets to inform better decisions.
3. **Personalization:** Delivers tailored experiences based on user behaviour.
4. **Accuracy:** Reduces errors in fields like healthcare and finance.
5. **24/7 Availability:** Operates continuously without breaks.
6. **Cost Savings:** Reduces operational costs by improving efficiency.
7. **Scalability:** Easily handles large volumes of tasks or data.
8. **Innovation:** Aids in generating creative solutions and ideas.
9. **Healthcare:** Enhances early diagnosis and treatment personalization.
10. **Security:** Improves real-time detection of threats like fraud or cyberattacks.

5. Limitations of AI

1. **Bias:** AI can reflect biases present in training data.
2. **Narrow Scope:** AI often struggles to generalize across different domains.
3. **High Costs:** Developing advanced AI systems requires significant investment.
4. **Ethical Concerns:** Raises issues related to surveillance and autonomous systems.
5. **Job Losses:** Automation can displace workers in certain sectors.
6. **Data Dependency:** Performance relies on high-quality data.
7. **Opacity:** Some AI models, like deep learning, lack explainability.
8. **Security Risks:** Vulnerable to adversarial attacks that deceive AI systems.
9. **Limited Creativity:** Lacks genuine human creativity and intuition.
10. **Regulatory Challenges:** Rapid AI development complicates legal frameworks and accountability.

6. Classification of AI

6.1. Based on Functionality:

1. **Narrow AI (Weak AI):** AI systems that are designed and trained to perform a specific task or a narrow range of tasks. These systems do not possess general intelligence and

cannot perform tasks outside their designated scope. **Examples:** Voice assistants like Siri and Alexa, Recommendation systems on Netflix or Amazon, Facial recognition software.

2. **General AI (Strong AI):** A hypothetical AI that possesses the ability to understand, learn, and apply knowledge across a wide range of tasks, similar to human intelligence. General AI can perform any intellectual task that a human can.
3. **Superintelligent AI:** A hypothetical AI that surpasses human intelligence in all aspects, including creativity, problem-solving, and decision-making. This AI would have the capability to outperform humans in virtually every field.

6.2. Based on Capabilities:

1. **Reactive Machines:** The most basic type of AI, these systems can only react to specific inputs with pre-programmed responses. They do not have memory or the ability to learn from past experiences. **Examples:** IBM's Deep Blue chess-playing computer, which could calculate millions of moves to play chess but had no understanding beyond that.
2. **Limited Memory:** AI systems that can use past experiences to inform current decisions, but their memory is not permanent. They can remember and use data for a short period of time to improve their actions. **Examples:** Self-driving cars that use data from the environment to make real-time decisions, like recognizing traffic patterns or obstacles.
3. **Theory of Mind:** An advanced type of AI that, in theory, would be able to understand emotions, beliefs, intentions, and thoughts of other beings. This level of AI would interact with humans in a way that is socially and emotionally aware. **Examples:** This type of AI is still in the research stage and has not been fully developed.
4. **Self-Aware AI:** The most advanced form of AI, which would possess self-awareness and consciousness. This AI would have its own perceptions, desires, and beliefs. This is still theoretical and not realised till date. They would be smarter and more capable than human mind **Examples:** Self-aware AI remains purely speculative and has not been achieved.

6.3. Based on Technology:

1. **Rule-Based AI:** AI systems that operate based on predefined rules and logic. These systems follow explicit instructions given by human programmers. **Examples:** Expert systems, like MYCIN, which was designed to diagnose bacterial infections.
2. **Machine Learning-Based AI:** AI systems that improve their performance over time by learning from data, without being explicitly programmed for every possible scenario.
Subcategories:
 - **Supervised Learning:** Trained on labelled data (e.g., spam detection).
 - **Unsupervised Learning:** Trained on unlabelled data (e.g., clustering algorithms).
 - **Reinforcement Learning:** Learns through trial and error, receiving rewards or penalties based on actions (e.g., AI in video games).
3. **Deep Learning-Based AI:** A subset of machine learning that uses neural networks with many layers (deep networks) to model complex patterns in data. Deep learning is particularly effective in tasks such as image and speech recognition. **Examples:** Google's

AlphaGo, which defeated a world champion in the game of Go, Autonomous vehicles using deep learning for object detection.

4. **Neural Networks:** AI systems modelled after the human brain, using interconnected nodes (neurons) to process information. They are the foundation of deep learning and are used for tasks like pattern recognition and predictive modelling. **Examples:** Image classification models, speech recognition systems, natural language processing.

6.4. Based on Applications:

1. **Artificial Narrow Intelligence (ANI):** AI that specializes in one area, capable of performing a single task extremely well. This is the most common form of AI currently in use. **Examples:** Virtual assistants, chatbots, recommendation engines.
2. **Artificial General Intelligence (AGI):** AI that can understand, learn, and apply intelligence across a wide range of tasks, much like a human. AGI would be capable of performing any intellectual task that a human can. **Examples:** AGI remains theoretical and has not been achieved.
3. **Artificial Superintelligence (ASI):** AI that surpasses human intelligence across all fields. This AI would be capable of outperforming humans in every aspect, from scientific discovery to artistic creativity. **Examples:** ASI is a speculative concept, often discussed in the context of future AI developments and ethical considerations.

7. Agent: An agent is an entity that perceives its environment through sensors and acts upon that environment through actuators. In AI, an agent can be a software program, a robot, or any system that can make decisions to achieve certain goals. **Types of Agents:**

1. **Learning Agents:** Improve their performance over time based on past experiences.
2. **Intelligent Agent:** is an autonomous entity that perceives its environment, makes decisions, and takes actions to achieve specific goals or objectives. It operates based on the principles of AI and is designed to function independently, adapt to changes, and improve its performance through learning and interaction.
3. **Simple Reflex Agents:** act solely based on the current percept, ignoring the history of precepts. They select actions based on a set of condition-action rules, often referred to as "if-then" rules. **Example:** A thermostat that turns the heater on when the temperature drops below a certain point and off when it rises above a threshold.

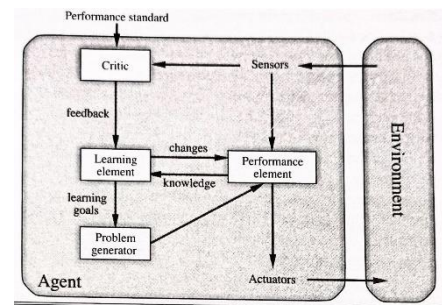


Figure 2.15 A general learning agent.

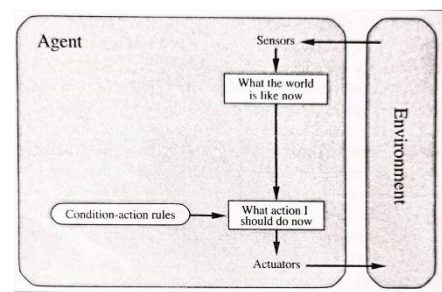


Figure 2.9 Schematic diagram of a simple reflex agent.

4. **Model-Based Reflex Agents:** maintain an internal model of the world, which helps them keep track of the parts of the environment that are not directly observable at the current moment. **Example:** A robot vacuum cleaner that builds a map of the room as it cleans, allowing it to avoid areas it has already cleaned.

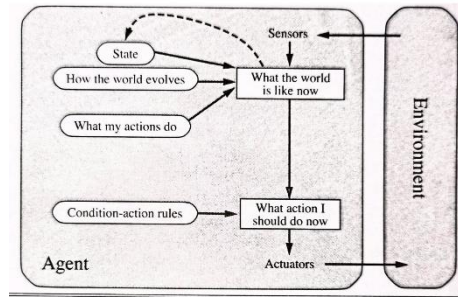


Figure 2.11 A model-based reflex agent.

5. **Goal-Based Agents:** Goal-based agents make decisions based not only on the current situation but also on the goals they are trying to achieve. These agents choose actions that lead them closer to achieving their goals. **Example:** A self-driving car that navigates toward a destination while considering the current.

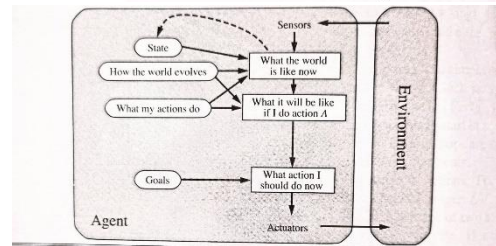


Figure 2.13 A model-based, goal-based agent. It keeps track of the world state as well as a set of goals it is trying to achieve, and chooses an action that will (eventually) lead to the achievement of its goals.

6. **Utility-Based Agents:** go beyond goal-based agents by not only aiming to achieve goals but also by considering the best way to achieve those goals. **Example:** An investment AI that not only aims to increase profit (goal) but also considers risk and time horizon to maximize expected returns (utility).

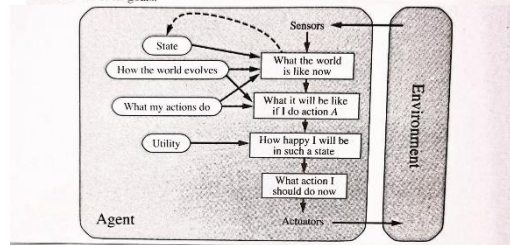


Figure 2.14 A model-based, utility-based agent. It uses a model of the world, along with a utility function that measures its preferences among states of the world. Then it chooses the action that leads to the best expected utility, where expected utility is computed by averaging over all possible outcome states, weighted by the probability of the outcome.

Feature	Simple Reflex Agents	Model-Based Reflex Agents	Goal-Based Agents	Utility-Based Agents
Basis of Operation	Operate on the current percept only.	Use an internal model to keep track of the world state.	Act to achieve defined goals.	Act to maximize a utility function.
Environment Interaction	React to stimuli with conditioned actions.	Update internal state from current state and action history.	Consider future consequences of actions.	Evaluate success based on a utility function.
Flexibility	Very limited; cannot handle new situations well.	More flexible, can handle unseen scenarios to an extent.	Flexible; can adapt actions to achieve goals in changing environments.	Highly flexible; aims for optimal performance.
Learning Ability	None; they do not learn from past actions.	Limited; can improve the model with new experiences.	Can learn better strategies to achieve goals.	Can adjust utility function based on outcomes.
Example	Thermostat controlling a heating system.	A car with anti-lock braking system.	Chess-playing AI considering future moves.	Investment AI maximizing portfolio return.

8. Environment: The environment refers to the external world in which the agent operates. It includes everything that the agent interacts with or perceives, and it can be anything from a physical space, like a room or a battlefield, to a virtual space, like a game world or the internet.

1. **Fully Observable vs. Partially Observable:** In a fully observable environment, the agent's sensors give it always access to the complete state of the environment. In a partially

observable environment, the agent has only partial information about the state. **Unobservable:** If the agent has no sensors at all then the environment is unobservable.

2. **Deterministic vs. Stochastic:** In a deterministic environment, the next state of the environment is completely determined by the current state and the action taken by the agent. In a stochastic environment, there is some level of unpredictability.
3. **Static vs. Dynamic:** A static environment remains unchanged except by the actions of the agent, while a dynamic environment can change on its own over time.
4. **Discrete vs. Continuous:** A discrete environment has a limited number of distinct states and actions, while a continuous environment has a range of possible states and actions.
5. **Single-Agent Environment:** In a single-agent environment, only one agent exists and interacts with the environment. **Example:** A chess-playing AI where the environment consists only of the chessboard and the agent makes decisions based on it. **Multi-Agent Environment:** In a multi-agent environment, multiple agents coexist and interact with the environment, as well as with each other. **Example:** A market simulation where multiple AI agents are competing to maximize their profit.
6. **Episodic Environment:** In an episodic environment, the agent's actions are divided into separate, independent episodes. Each episode consists of a single interaction with the environment. **Example:** Image classification tasks where the AI model classifies images independently of each other. **Sequential Environment:** In a sequential environment, the current decision can affect future states and decisions. The agent's actions have a long-term impact. **Example:** Playing a game of chess, where each move influences subsequent moves and the eventual outcome.
7. **Known Environment:** In a known environment, the agent has complete information about the environment, including the rules, states, and the effects of its actions. **Example:** A robot navigating a pre-mapped building where it knows the layout and possible obstacles in advance.
8. **Unknown Environment:** In an unknown environment, the agent has incomplete or no prior knowledge of the environment's dynamics. The agent must learn about the environment as it interacts with it. **Example:** A robot exploring an unfamiliar terrain where it must learn the layout and obstacles through exploration.

Task Environment	Observable	Agents	Deterministic	Episodic	Static	Discrete
Crossword puzzle	Fully	Single	Deterministic	Sequential	Static	Discrete
Chess with a clock	Fully	Multi	Deterministic	Sequential	Semi	Discrete
Poker	Partially	Multi	Stochastic	Sequential	Static	Discrete
Backgammon	Fully	Multi	Stochastic	Sequential	Static	Discrete
Taxi driving	Partially	Multi	Stochastic	Sequential	Dynamic	Continuous
Medical diagnosis	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
Image analysis	Fully	Single	Deterministic	Episodic	Semi	Continuous
Part-picking robot	Partially	Single	Stochastic	Episodic	Dynamic	Continuous
Refinery controller	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
Interactive English tutor	Partially	Multi	Stochastic	Sequential	Dynamic	Discrete

Figure 2.6 Examples of task environments and their characteristics.

9. Key Concepts

1. **Percept:** A percept is the input that an agent receives from its environment at any given time. It is essentially the agent's perception or observation of the environment. **Example:** For a robot, a percept might be the data from its cameras and sensors, such as detecting an obstacle in its path. For a software agent in a chess game, a percept could be the current configuration of the chessboard.
2. **Percept Sequence:** The percept sequence is the complete history of everything the agent has perceived. It includes all percepts that the agent has received from the environment over time, from the initial state to the current state. **Example:** In the case of a robot vacuum cleaner, the percept sequence could include a record of all the obstacles it has encountered and the positions it has cleaned so far.
3. **Agent Program:** The agent program is the software or algorithm that runs on the agent and determines its behaviour. It takes the current percept (and possibly the percept sequence) as input and outputs an action for the agent to take. **Example:** For a chess-playing AI, the agent program might use a decision tree or a search algorithm to determine the best move based on the current board configuration.
4. **Agent Function:** The agent function is a mathematical mapping from the percept sequence to an action. It is an abstract concept that describes what the agent does for any given percept sequence. The agent program is the practical implementation of this function. **Example:** If an agent's task is to navigate a maze, the agent function would define how the agent chooses its actions (e.g., move forward, turn left) based on the sequence of percepts it has received (e.g., walls detected, path ahead clear).

10. PEAS (Performance measure, Environment, Actuators, Sensor)

PEAS is used to describe the components of an intelligent agent's environment and its functioning. It stands for Performance measure, Environment, Actuators, and Sensors. These components are crucial when designing an intelligent agent, as they help in defining the agent's interaction with the world.

1. **Performance Measure:** defines the criteria by which the agent's success is evaluated. It specifies what counts as a successful outcome in the given environment. **Example:** In a self-driving car, the performance measure could be the safety of the passengers, fuel efficiency, or minimizing travel time.
2. **Environment:** is the world in which the agent operates. It consists of everything that surrounds the agent and influences its decisions. **Example:** For a vacuum cleaner robot, the environment would be the rooms, dirt on the floor, and obstacles.
3. **Actuators:** are the mechanisms through which the agent acts upon its environment. They allow the agent to perform actions. **Example:** In a robot, actuators include wheels, arms, or any mechanism used for movement or manipulation of objects.
4. **Sensors:** Sensors allow the agent to **perceive** its environment by collecting data about the state of the world. The data from sensors is used by the agent to make informed decisions. **Example:** For a self-driving car, sensors include cameras, LiDAR, radar, and GPS systems.

Different Agents	Performance Measure	Environment	Actuators	Sensors
Self - Driving Car	Safety, minimizing travel time, fuel efficiency	Roads, other vehicles, pedestrians, traffic lights	Steering, accelerator, brakes	Cameras, radar, GPS, LiDAR
Vacuum Cleaner Robot	Cleanliness of the floor, time taken to clean	The floor, furniture, walls, and dirt	Wheels, vacuum motor, brushes	Dirt sensors, bump sensors, cameras for navigation
Chess AI	Winning the game or maximizing score	The chessboard and the opponent's moves	Moving the pieces on the board	Perceiving the opponent's moves, board state
Smart Thermostat	Maintaining desired room temperature efficiently, energy savings	Indoor room, temperature, time of day, external weather conditions	Heating, ventilation, air conditioning (HVAC) systems	Temperature sensors, humidity sensors, occupancy sensors
Autonomous Drone	Safe navigation, avoiding obstacles, completing tasks	Airspace, obstacles (buildings, trees), weather conditions	Propellers, camera tilt, landing gear	GPS, cameras, altimeters, LiDAR, accelerometers
Speech Recognition System	Accuracy of speech recognition, response relevance, user satisfaction	User's speech, background noise, language context	Voice responses, text display, executing user commands	Microphone, ambient noise detection, voice recognition software
Warehouse Robot/ Part-picking robot	Efficient item picking and placement, reducing errors, timely task completion	Warehouse layout, shelves, items to pick, obstacles	Robot arms for picking items, wheels for movement	Cameras, barcode scanners, proximity sensors, weight sensors
Online Shopping Recommendation System	User engagement, sales conversion, relevance of recommendations	User browsing data, purchase history, product catalog	Displaying product recommendations, emails, notifications	User clicks, searches, purchase data, browsing behavior
Medical Diagnosis System	Accuracy of diagnosis, treatment recommendations, patient outcomes	Patient medical history, symptoms, test results	Displaying diagnosis or treatment options to doctors, suggesting medications	Input from doctors, medical records, patient data (symptoms, tests)
Automated Trading System	Profit maximization,	Stock market data, financial news,	Buying/selling financial assets,	Market data feed, financial analysis tools, news inputs

	risk management, transaction speed.	economic indicators	generating trade signals	
Satellite image analysis system	Correct image categorization.	Downlink from orbiting satellite.	Display of scene categorization.	Color pixel arrays.
Interactive English tutor	Student's score on test.	Set of students, testing agency.	Display of exercise, suggestions, corrections	Keyboard entry

Importance of PEAS

1. **Defines Agent Behaviour:** PEAS helps specify how an AI agent should act in its environment.
2. **Performance Measurement:** Establishes clear criteria to evaluate the agent's success.
3. **Environment Understanding:** Describes the surroundings in which the agent operates.
4. **Action Specification:** Identifies the actions the agent can perform using actuators.
5. **Input Handling:** Clarifies how sensors help the agent perceive the environment.
6. **Task Clarification:** Outlines the task the agent is expected to perform effectively.
7. **Structured Problem Solving:** Provides a systematic way to design intelligent agents.
8. **Adaptability:** Ensures the agent can adjust its actions based on sensor inputs.
9. **Goal Achievement:** PEAS ensures the agent is aligned with achieving its objectives.
10. **Efficiency:** Optimizes how an agent interacts with its environment for better performance.

11. Uncertainty in Artificial Intelligence

Uncertainty in AI refers to the situation where an AI agent cannot predict outcomes with complete accuracy due to incomplete, ambiguous, or noisy information about the environment. It involves the inability to be certain about the state of the world or the results of actions.

Causes of Uncertainty:

1. **Incomplete Information:** The agent does not have access to all relevant data or facts about the environment.
2. **Ambiguity:** The available data might have multiple interpretations (e.g., vague descriptions).
3. **Stochastic (Random) Nature:** The environment may behave unpredictably or involve random events.
4. **Sensor Noise:** Sensors used by the agent to gather data can be faulty or noisy, providing inaccurate information.
5. **Dynamic Environments:** Changes in the environment over time may introduce uncertainty if the agent's knowledge is not updated in real-time.

Sources of Uncertainty:

1. **Perception:** Errors or noise in the data collected by sensors (e.g., vision systems, microphones).
2. **External Factors:** Unpredictable or uncontrollable events in the environment (e.g., weather changes, human intervention).
3. **Inherent Complexity:** The world itself may have elements that are inherently unpredictable, such as market fluctuations or biological processes.
4. **Lack of Knowledge:** The agent may lack critical knowledge about the world or the effects of its actions.
5. **Non-deterministic Actions:** Certain actions may have probabilistic outcomes rather than a guaranteed result (e.g., rolling a dice).

Mechanisms for Handling Uncertainty:

1. **Probability Theory:** Uses probabilistic models to represent uncertainty in knowledge and outcomes. Bayesian networks and probabilistic reasoning are commonly used to update beliefs based on new evidence. **Example:** Predicting weather where the likelihood of rain is 60%, based on historical data and current observations.
2. **Fuzzy Logic:** Fuzzy logic deals with vagueness and ambiguity by allowing degrees of truth rather than binary true/false outcomes. It is useful when the problem involves imprecise or vague data. **Example:** An AI controlling a thermostat can use fuzzy logic to adjust temperature based on subjective terms like “slightly cold” or “very warm.”
3. **Belief Networks (Bayesian Networks):** A belief network is a graphical model that represents the dependencies between variables using probability. It helps in reasoning under uncertainty by updating beliefs when new evidence is observed. **Example:** A medical diagnosis system uses a Bayesian network to update the likelihood of a disease given new test results.
4. **Decision Theory:** Combines probability (expected outcomes) with utility theory(cost/benefit) to make rational decisions under uncertainty. **Example:** In an AI-driven stock trading system, decision theory helps balance risk and reward by selecting trades with the highest expected utility.
5. **Monte Carlo Simulation:** A computational technique that relies on repeated random sampling to estimate the probabilities of different outcomes. It is useful for systems where calculating exact probabilities is complex. **Example:** Monte Carlo methods are used in games like poker to simulate possible future game states and calculate the best move.
6. **Non-probabilistic Methods (e.g., Rule-based Systems):** Some systems handle uncertainty using heuristic or rule-based approaches without relying on probabilities. These systems may use logical rules to infer outcomes based on known conditions. **Example:** An expert system for medical diagnosis may use a set of rules like "If symptom X is present, then disease Y is likely."